

RALPH JOSEPH DSOUZA

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OBJECTIVE

Computer Engineering undergraduate experienced in building software end-to-end: full-stack apps (Python/FastAPI, React/TypeScript), classical ML systems with explainability, and LLM-integrated tools (Gemini/GPT). Seeking a Trainee Software Engineer role to build production-grade, scalable SaaS features.

EDUCATION

Fr. Conceicao Rodrigues College of Engineering, Mumbai B.E. Computer Engineering (Pursuing) — Graduating 2027	2023 – 2027
Podar International School, Mumbai ISC Higher Secondary — 87.5%	2021 – 2023

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript (ES6+), C
Backend & APIs: FastAPI, Flask, RESTful API Design, MongoDB, Streamlit
Frontend: React.js, TypeScript, HTML5, CSS3, Apache ECharts
ML/AI: Scikit-learn, XGBoost, LightGBM, PyTorch, SHAP, Optuna, imbalanced-learn (SMOTE-Tomek), OpenCV, Reinforcement Learning (PPO, Gymnasium), Flower (Federated Learning)
LLM/GenAI: Gemini API, GPT API, Prompt Engineering, AI-Assisted Development
Tools & DevOps: Git, GitHub, GitHub Actions (CI/CD), Vercel, Railway, Linux

EXPERIENCE

Web Development Intern <i>MeshNet Electronics LLP, Mumbai</i>	Dec 2025 – Present
Built a real-time telemetry dashboard from scratch (JavaScript, Apache ECharts) consuming live REST API sensor data, with frontend data normalization and error-state handling.	
AI Generalist Intern <i>HumbleWalking, Mumbai</i>	2 Months
Applied AI-assisted workflows for SEO/AEO optimization and structured metadata on a live web platform, using AI tools to accelerate content refinement and iterative product improvements.	
Lead <i>Game Developers Association, FRCRCE, Mumbai</i>	Jul 2025 – Present
Lead operations, event planning, and cross-functional coordination for the college's Game Developers Association.	

MACHINE LEARNING & AI PROJECTS

Credit Risk Intelligence Platform <i>Python, Scikit-learn, XGBoost, LightGBM, SHAP, Optuna</i>
- Built an end-to-end credit-risk platform training and comparing 4 classifiers (Logistic Regression, XGBoost, LightGBM, Random Forest) on 50K loan records, reaching 0.996 AUC-ROC on the best model. - Handled class imbalance with SMOTE-Tomek and tuned hyperparameters via Optuna Bayesian search; built a SHAP-based explainability dashboard in Streamlit for per-prediction transparency, plus stress-testing and survival-analysis modules.
QuestionForge AI <i>FastAPI, React, TypeScript, Gemini/GPT API</i>
- Built a full-stack LLM-powered assessment platform from scratch that generates application-based question papers and analyzes them against Bloom's Taxonomy for difficulty and readiness metrics. - Implemented automated evaluation of scanned PDF answer sheets with hybrid AI logic — LLM-based scoring with local fallback heuristics for offline/quota-limited scenarios.
Federated Review Analyzer <i>PyTorch, Flower, Streamlit</i>
- Built a privacy-preserving federated learning system training an LSTM sentiment classifier across simulated heterogeneous retail clients without sharing raw data, using the Flower framework for asynchronous FedAvg orchestration. - Developed an interactive Streamlit UI for live training visualization and real-time inference on the aggregated global model.
Lane Detection with Computer Vision & Reinforcement Learning <i>OpenCV, Gymnasium, PPO</i>
- Implemented lane detection and turn recognition using Canny edge detection and Hough Transform on a simulated driving environment. - Trained a PPO reinforcement learning agent (Gymnasium CarRacing) for smooth, controlled lane-changing behavior.

OTHER PROJECTS

- IoT Monitoring Dashboard** | Python, Flask, JavaScript, Apache ECharts — Real-time sensor dashboard with anomaly alerts from a Flask REST backend.
- E-Commerce Store** | JavaScript, REST APIs — Product listing, cart, and checkout flow with API-driven architecture.

CERTIFICATIONS

100 Days of Code (Udemy) · Python: Beginner to Mastery (Zero to Mastery) · Java Programming (Udemy) · C Programming for Beginners (Udemy)